Bizocity Score at AT&T

One very interesting application of data mining to the task of finding patterns corresponding to predefined customer segments is the system that AT&T Long Distance uses to decide whether a phone is likely to be used for business purposes. AT&T views anyone in the United States who has a phone and is not already a customer as a potential customer. For marketing purposes, they have long maintained a list of phone numbers called the Universe List. This is as complete as possible a list of U.S. phone numbers for both AT&T and non-AT&T customers flagged as either business or residence. The original method of obtaining non-AT&T customers was to buy directories from local phone companies, and search for numbers that were not on the AT&T customer list. This was both costly and unreliable and likely to become more so as the companies supplying the directories competed more and more directly with AT&T. The original way of determining whether a number was a home or business was to call and ask. AT&T Long distance had separate services for business and home customers.

Corina Cortes and Daryl Pregiborn, researchers at Bell Labs (then a part of AT&T) came up with a better way. AT&T, like other phone companies, collects call detail data on every call that traverses its network (they are legally mandated to keep this information for a certain period of time). Many of these calls are either made or received by non-customers. The telephone numbers of non-customers appear in the call detail data when they dial AT&T 800 numbers and when they receive calls from AT&T customers. These records can be analyzed and scored for likelihood to be businesses based on a statistical model of businesslike behavior derived from data generated by known businesses. This score, which AT&T calls "bizocity," is to determine which services should be marketed to the prospects.

Every telephone number is scored every day. AT&T's switches process several hundred million calls each day, representing about 65 million distinct phone numbers. Over the course of a month, they see over 300 million distinct phone numbers. Each of those numbers is given a small profile that includes the number of days since the number was last seen, the average daily minutes of use, the average time between appearances of the number on the network, and the bizocity score.

The bizocity score is generated by a regression model that takes into account the length of calls made and received by the number, the time of day that calling peaks, and the proportion of calls the number makes to known businesses. Each day's new data adjusts the score. In practice the score is a weighted average over time with the most recent data counting the most.

Bizocity can be combined with other information in order to address particular business segments. One segment of particular interest in the past is home businesses. These are often not recognized as businesses even by the local phone company that issued the number. A phone number with high bizocity that is at a residential address or one that has been flagged as residential by the local phone company is a good candidate for services aimed at people who work at home.